

Campaign audit R397–R426

independent re-computation of seven load-bearing targets; no round verdict
flips

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Abstract

Rounds 397–426 produced an intertwiner, a quasi-definite mass ratio, a kernel-Loewner comparison $\|b\|^2 \leq S$, a floor $R_\infty \approx 0.030$, and a four-coordinate death chart. This note is an *adversarial* re-verification on independent code paths (r367 $A_0 + \frac{1}{2}I$ for R ; dense Chebyshev–Hankel for C ; full-carrier product dual; KKT $\|\mathfrak{I}_0\|$; Woodbury-sch). Search raster: truncation/depth mismatch, signed vs occupied chain mix-up, shared-builder bugs, pin circularity, float64 on 10^{-8} razors. Ausgang: **0 BUG / 2 SUSPECT / 5 CLEAN**. No round verdict of R397–R426 changes. No RH claim.

Status. Adversarial audit closed. Known artefact classes of r395 ($n_{\text{ref}} = 60$) and r381/r390 (signed vs occupied Fejér) are *absent* from the 17 campaign probes. Two process findings remain (floor abscissa; disclosed $k = 8$ pin reuse). No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The audit

Lemma 1 (campaign numbers). *The seven load-bearing campaign numbers of R407–R425 are either independently reproduced or flagged as process-suspect; none is a silent bug of the r395/r381 class.*

Verdict: **CLOSED** as an audit, not as a theorem of the wall. Table 1 is the record.

2 Intertwiner

Proposition 1 (independent Möbius). *On $k_z = 9, 17, 12$, the r367 block $R_{N-3} = A_0 + \frac{1}{2}I$ agrees with $R = C(I + C)^{-1}$ built from a dense Chebyshev–Hankel Gram on Y (not the campaign Cholesky), and*

$$\text{ind}_-(R - \tfrac{1}{2}I) = \#\{\lambda(C) < 1\}.$$

Over $\mathbb{Q}$: $C = \text{diag}(2, 1/2)$ gives $R = \text{diag}(2/3, 1/3)$ and both counts equal 1. On w9: $C_{\min} = 0.857119$ (pins 0.85712 / 0.857119), formula $3.38 \cdot 10^{-12}$. On kz17: $C_{\min} = 0.279842$, one negative half-mode. On kz12: $C_{\min} = 1.000074$, dictionary 0 = 0 (PD bulk). mpmath dps-40 vs float64 Hankel at $n = 12$: relative $\leq 1.5 \cdot 10^{-14}$.

kz42 ($N_w = 2472$) was *not* Hankel-rebuilt (native depth is the r395 class). The PD razor of r411 is audited via $\kappa(C)$ and an independent kz12 window, not by native Hankel at $N = 2472$.

Table 1: Verdict per target. Independent re-computation, then CLEAN / BUG / SUSPECT.

T	Verdict	Independent numbers
1	CLEAN	$R = C(I + C)^{-1}$ on $k_z = 9, 17, 12$; $\ R_f - R\ _{w9} = 3.38 \cdot 10^{-12}$; $\text{ind}_- = \#\{\lambda(C) < 1\} (1, 1, 0)$.
2	CLEAN	$\sum_X u^\vee / \sum_Y u^\vee = 1.357130$ from one full-carrier dual; subset- P' would have given 1.525.
3	CLEAN	$\ \mathfrak{T}_0\ _{w9} = 1.080138437324$; $\kappa(C) = 2.086 \cdot 10^3$ so $4.57 \cdot 10^{-8}$ is $56\times$ eigvalsh noise; r409 1.00000 is display rounding.
4	CLEAN	$K^{\mu-\nu} \succeq K^\mu$ on truncated PD Grams $n = 16$ ($\text{evmin} \sim -10^{-9}$); indefinite $\mathbb{Q}$ -toy blocks inversion monotonicity.
5	SUSPECT	vs k : M1 $R_\infty = +0.02983$ (r421 +0.02982); vs N : winner M3 ($N = 5690$ at $k = 8$, $N = 1433$ at $k = 9$); $k = 8$ point is a builder pin.
6	CLEAN	6/6 dead: $q_N > 1 \Leftrightarrow Z_{\text{loc}} _{\text{term}} \geq \sqrt{5/7} \Leftrightarrow \text{sch} > 0$; live $\chi 3-9$ does not fire; razor-overlap separates living from dead.
7	SUSPECT	no $n_{\text{ref}}/\text{float32}$; $k = 8$ pin circularity r421→r425 disclosed; razor C_{min} in float64 eigvalsh .

3 QD mass

Proposition 2 (one dual). *The r413 ratio 1.357 is the restriction of a single full-carrier dual $u^\vee = 1/(u \cdot P_\Xi^2)$ with P_Ξ on the whole grid, then split to X and Y .*

Route A (log-shifted product) and route B (r356 κ -formula) are proportional at $3.9 \cdot 10^{-12}$. A subset- P' computed separately on X and on Y yields 1.525 — that is the X/Y-norm bug; the campaign does not use it. Signed u produces $|Y|$ negative duals; the campaign uses $|u|$.

Impact if the subset bug had been live: both the QD-refutation and the mass asymmetry would move. It is not live.

4 $\|\mathfrak{T}_0\|$ and float64 razors

Independent KKT Chebyshev interpolant: $\|\mathfrak{T}_0\|_{w9} = 1.080138437324$, matching r411/r413/r416 pins at 1.080138437 and $1/\sqrt{C_{\text{min}}}$ at $5 \cdot 10^{-12}$. The printed r409 band $[1.05, 1.12]$ contains the tight pin.

The r409 PD print $\|\mathfrak{T}_0\| = 1.00000$ is rounding of $1/\sqrt{1 + 4.565 \cdot 10^{-8}} = 0.999999977$. The gap $4.57 \cdot 10^{-8}$ sits $56\times$ above $\varepsilon_{\text{mach}} \cdot \kappa(C) \cdot C_{\text{max}}$ with $\kappa(C) = 2.086 \cdot 10^3$ on w9 (Hankel $\text{cond}(G) \sim 10^7$ is *not* the eig-scale of C). Independent kz12: $C_{\text{min}} - 1 = 7.36 \cdot 10^{-5}$, $\kappa(C) = 29$, noise $\sim 10^{-13}$.

Campaign razors run in float64 **eigvalsh**. At the measured $\kappa(C)$ this is adequate; a $\kappa(C) \gtrsim 10^8$ window would make a 10^{-8} gap indistinguishable from noise. That is T7's precision ward, not a flipped r411 verdict.

5 Kernel Loewner

If both Gram forms are positive definite on $P_{<N}$ and $\nu \geq 0$, then $\langle p, p \rangle_\mu \geq \langle p, p \rangle_{\mu-\nu}$ and inversion monotonicity gives $K^{\mu-\nu} \succeq K^\mu$, hence $\|b\|^2 \leq S$. Over $\mathbb{Q}$: $\mu = (2, 1)$ at $(0, 1)$, $\nu = (1)$ at 1, constants, $K^\mu = 1/3 \leq K^{\mu-\nu} = 1/2$.

If the tilde form is indefinite the hypothesis is false: $\mu = (3, 1)$ at $(0, 1)$, $\nu = (2)$ at 1, depth 2, $G(\mu - \nu)$ has $\lambda_{\min} = -1.30$; $K_t - K_\mu$ has $\lambda_{\min} = -2$. The campaign states the tilde chain is definite; the implication then holds. Truncated PD Grams $n = 16$ on $k_z = 9, 17, 12$: $K \succeq K_\mu$ with $\lambda_{\min}(K - K_\mu) \sim -10^{-9}$ (float64 residual).

6 Floor fit

Re-fit of the published selected sequence ($k = 5..9$, including the $k = 8$ builder pin $R = 0.04763$, $N = 5690$):

abscissa	winner	R_∞ (M1)	note
k	M1	+0.02983	reproduces r421
N	M3	+0.02840	N not monotone
S	M2	—	log-law still dead
2^k	M1	+0.034	—

N at $k = 8$ is 5690; at $k = 9$ it is 1433. Fitting against k conceals that inversion. r421 already typed the floor as a census, not a theorem (COFINAL_OPEN), and discarded the N -fit because N is illegal as a cofinal abscissa. The EXT-VAC band $[0.0285, 0.0373]$ is the published r421 interval (gate bar in the probe is $[0.025, 0.040]$); the k -floor sits inside it. The band was *not* independently rebuilt from EXT windows in this audit (native $N \geq 1119$).

Impact: no round verdict flips. $\|b\|^2 \leq S$ and the intertwiner do not depend on $R_\infty \approx 0.030$. The sentence “the cofinal law is a positive floor 0.030” is weaker than the campaign heading suggested.

7 Dead χ coordinates

On all six dead windows ($\chi_3 \in \{15, 19, 23, 33, 39\}$, $\chi_4 = \{20\}$):

$$q_N > 1 \iff |Z_{\text{loc}}|_{\text{term}} \geq \sqrt{5/7} \iff \text{sch} > 0.$$

The middle equivalence is definitional ($|Z_{\text{loc}}|_{\text{term}}^2 = q_N \cdot 5/7$). Live control χ_3 -9: $q_N = 0.247$, $\text{sch} = -0.065$, overlap $6.9 \cdot 10^{-7}$. Dead overlaps $8 \cdot 10^{-6}$ to 10^{-3} (not uniformly 10^{-4}). $C_{\min} - 1$ is *not* a death coordinate: χ_3 -23, χ_3 -33, χ_4 -20 have $C_{\min} > 1$ and still die at the edge (r401).

8 Code hunt

Seventeen campaign probes, grep-backed: no nref=INT, no float32, no seeded result paths. $k = 8$ builder pin is *disclosed* in r421 and reused in r425 (K8_PIN, “not re-gated”). X/Y masses come from one dual_split. Razor $C_{\min}$ is float64 eigvalsh (T3: $56\times$ noise on w9).

9 Impact

No BUG. T5 and T7 are process findings: the floor is abscissa-dependent and one fit-point is a pin; razors live in float64 at a still-comfortable margin. R407 intertwiner, r413 QD ratio, r411 $\|\mathfrak{T}_0\|$, r425 Loewner, and the q_N/sch death chart stand.

10 What is a theorem

Nothing in this note is a theorem of the wall. It is an audit of previously sealed finite identities and censuses. No RH claim.